

Storage Devices

Lesson 10 — Magnetic, Optical, Flash and Cloud

JKSSB Computer Awareness

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Storage in the Memory Hierarchy

- ▶ **Primary storage:** directly accessible by the CPU; holds data and instructions currently in use (RAM, ROM, cache).
- ▶ **Secondary storage:** non-volatile; keeps data when the power is off (hard disk, SSD, optical disc, USB drive).
- ▶ **Tertiary storage:** archival level, often robotic tape libraries that are loaded on demand.

Anchor point

The CPU works with main memory; secondary storage feeds main memory through the operating system.

Magnetic Storage: Hard Disk Drive

- ▶ Data is stored magnetically on **rotating platters**.
- ▶ A **read/write head** on an actuator arm moves over the platter surface to read or write.
- ▶ It is a mechanical device: the platters spin and the head moves.
- ▶ Large capacity at low cost per gigabyte.

Remember

An HDD is magnetic and mechanical — it has moving parts.

Magnetic Tape: Sequential Storage

- ▶ Data is stored magnetically on a long tape in a reel or cartridge.
- ▶ The tape is read or written in order, one position after another.
- ▶ This is **sequential access**, so reaching a distant record takes longer.
- ▶ Widely used for backup and long-term archival rather than day-to-day access.

Tape is cheap and dense, but slow for random lookups.

Optical Storage

- ▶ Data is stored as microscopic **pits and lands** and read by a **laser**.
- ▶ Common forms: **CD**, **DVD**, and **Blu-ray** disc.
- ▶ Optical media are portable, removable, and non-volatile.
- ▶ Blu-ray uses a shorter-wavelength laser and tighter tracks, so it holds more than a DVD.

Capacity cue

CD $\approx$ 700 MB; DVD $\approx$ 4.7 GB (single layer); Blu-ray $\approx$ 25 GB (single layer).

Optical Media Capacities

Medium	Approximate capacity (single layer)
CD	700 MB
DVD	4.7 GB
Blu-ray	25 GB

Order from smallest to largest: $\text{CD} < \text{DVD} < \text{Blu-ray}$.

Flash and Solid-State Storage

- ▶ **Flash memory** stores data in semiconductor cells with **no moving parts**.
- ▶ Examples: **SSD**, **USB flash drive**, and **memory card**.
- ▶ Because nothing spins, flash is faster and more shock-resistant than a mechanical drive.
- ▶ It is non-volatile: data remains when the power is off.

High-yield fact

Among these devices, the **SSD** has the highest transfer rate (KB PYQ-27).

Cloud Storage

- ▶ Data is stored on **remote servers** and accessed over the Internet.
- ▶ The provider manages the hardware; the user reaches the files through a network connection.
- ▶ It still ultimately rests on physical secondary storage in a data centre.
- ▶ Useful for backup, sharing, and access from many devices.

Cloud storage is remote secondary storage, not main memory.

HDD versus SSD

Hard disk drive

- ▶ Magnetic storage
- ▶ Mechanical, moving parts
- ▶ Rotating platters and a moving head
- ▶ Usually slower transfer rate
- ▶ More vulnerable to shock

Solid-state drive

- ▶ Flash storage
- ▶ No moving parts
- ▶ Semiconductor cells
- ▶ Higher transfer rate (fastest here)
- ▶ More shock-resistant

Random Access versus Sequential Access

- ▶ **Random (direct) access:** any location can be reached in approximately equal time. RAM and hard disks behave this way.
- ▶ **Sequential access:** locations are reached in order; the time depends on how far the target is from the current position. Magnetic tape is sequential.
- ▶ This distinction explains why tape is poor for direct lookups but fine for ordered backup.

Quick check

Memory and disks are approximately random access; tape is sequential.

Non-Volatility of Storage

- ▶ **Non-volatile** means the contents survive when the power is switched off.
- ▶ Primary storage such as RAM is **volatile**: its contents are lost without power.
- ▶ Secondary storage — HDD, SSD, optical, USB — is non-volatile, so files persist after shutdown.
- ▶ Do not say RAM is non-volatile; do not say storage is volatile.

One-line rule

RAM is volatile; storage is non-volatile.

Office Desktop Example (EX-01)

- ▶ An office desktop uses an **SSD** as the system drive, so the operating system boots quickly.
- ▶ A large **HDD** holds bulk documents and media at lower cost.
- ▶ A **DVD drive** and **USB flash drive** move files between machines.
- ▶ Important files are also copied to **cloud storage** for remote backup.

Fast flash for daily work; magnetic capacity for bulk; optical and USB for transfer; cloud for off-site backup.

Exam Traps to Avoid

- ▶ The **SSD** has the highest transfer rate, not the HDD (KB PYQ-27).
- ▶ An HDD is **magnetic and mechanical**; an SSD is **flash with no moving parts**.
- ▶ Optical capacities run $CD < DVD < \text{Blu-ray}$.
- ▶ RAM is **volatile** but storage is **non-volatile**.

Exam method

Match the storage type to its technology and its access method before answering.

Lesson 10: Recall

- ▶ Primary storage is CPU-accessible; secondary storage is non-volatile; tertiary storage is archival.
- ▶ Magnetic: HDD (rotating platters, moving head) and tape (sequential).
- ▶ Optical: CD $\approx$ 700 MB, DVD $\approx$ 4.7 GB, Blu-ray $\approx$ 25 GB (single layer).
- ▶ Flash: SSD, USB drive, memory card — no moving parts; the SSD has the highest transfer rate.
- ▶ RAM and disks are approximately random access; tape is sequential.